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MILITARY SPECIFICATION

COMPRESSORS, HERMETIC, AIR CONDITIONING, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers the requirements for 50/60-Hz and 400-Hz, electric-motor-driven hermetic compressors for use with air conditioni equipment using refrigerant -22 (monochlorodifluoromethane) and as specified in MIL-A-38339.

1.2 Classification. Compressors shall be of the following classes and types, as specified (see 6.2):

Class 1	- 7,000 BTU/hr nominal capacity
Type I	- 115-V, 50/60-Hz, single-phase
Type II	- 208-V, 400-Hz, 3-phase
Class 2	- 11,000 BTU/hr nominal capacity
Type I	- 115-V, 50/60-Hz, single-phase
Type II	- 230-V, 50/60-Hz, single-phase
Type III	- 208-V, 50/60-Hz, 3-phase
Type IV	- 208-V, 400-Hz, 3-phase
Class 3	- 20,000 BTU/hr nominal capacity
Type I	- 230-V, 50/60-Hz, single-phase
Type II	- 208-V, 50/60-Hz, 3-phase
Type III	- 208-V, 400-Hz, 3-phase
Class 4	- 40,000 BTU/hr nominal capacity
Type I	- 208-V, 50/60-Hz, 3-phase
Type II	- 208-V, 400-Hz, 3-phase
Class 5	- 60,000 BTU/hr nominal capacity
Type I	- 208-V, 50/60-Hz, 3-phase
Type II	- 208-V, 400-Hz, 3-phase

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-A-38339	Air Conditioners, Lightweight, Compact, Military, General Requirements For

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS3102	Connector, Receptacle, Electric, Box Mounting

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Heating, Refrigeration, and Air Conditioning Engineers, Incorporated

ASHRAE Standard 23	Methods of Testing and Rating Refrigerant Compressors

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(Application for copies should be addressed to the American Society of Heating, Refrigeration, and Air Conditioning Engineers, Incorporated, 345 East 47th Street, New York, New York 10017.)

National Electrical Manufacturers Association

Standard MG-1 Standards For Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

Underwriters' Laboratories, Incorporated

UL Std No. 303 Condensing Units, Refrigeration

(Application for copies should be addressed to the Underwriters' Laboratories, Incorporated, 161 Sixth Avenue, New York, New York 10013; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Materials. Exposed metal surfaces of the compressors shall be of the corrosion-resistant type or treated to resist corrosion due to atmospheric conditions likely to be met in storage and normal service and shall be suitable for the purpose. Materials shall be free from defects and imperfections that might affect the serviceability of the finished compressors.

3.3 Design

3.3.1 Reliability. The compressors shall each have a mean-time-between-failure (MTBF) of not less than 1,300 hours at a confidence factor of 0.90 (see 6.3.1 and 6.3.2).

3.3.2 Operating life. The compressors shall be capable of operating for 2,740 hours without loss of efficiency during that period.

3.3.3 Tilted position operation. The compressors shall be capable of operation without failure when tilted 10 degrees from vertical in any direction.

3.3.4 Refrigerant. Compressors shall be suitable for operation with refrigerant -22 (monochlorodifluoromethane).

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3.3.5 Motor cooling. The compressor motor shall be refrigerant-, air-, or oil-cooled, or any combination of the three.

3.3.6 Rupture pressure. The compressors shall be designed to withstand pressures up to 675 psig in both the suction and discharge cavities without rupture.

3.3.7 Compressor oil. The compressors shall be filled with lubricating oil to the level recommended by the manufacturer. The compressors shall be supplied with a grade and type of lubricating oil as recommended by the compressor manufacturer.

3.3.8 Housing. The compressor and compressor motor shall be enclosed in the same housing with the mounting hardware and no external shaft or shaft seal, and with the motor running in a refrigerant atmosphere. The housing may be of either the fully welded or brazed type, or of the service sealed type. In the fully welded or brazed type, the housing shall be permanently sealed and shall not be provided with means of access for servicing in the field. In the service-sealed type, the housing shall be provided with some means of access for servicing internal parts.

3.4 Performance

3.4.1 Operation. The compressors shall be capable of continuous operation under the conditions specified in table I.

TABLE I. Operating Conditions

Saturated Discharge Temp. (^{o}F)	Saturated Suction Temp. (^{o}F)	Ambient Temp. (^{o}F)
160	40 to 60	160
150	30 to 60	150
140	30 to 60	140
130	30 to 60	130
120	30 to 40	120
60	-20 to 20	-20 to 60

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3.4.1.1 Classes 1, 2, and 3 compressors shall be capable of operation in ambient temperatures down to 0°F. Classes 4 and 5 compressors shall be capable of operation in ambient temperatures down to -25°F. These requirements shall be obtained with the compressor installed in the appropriate air conditioner (MIL-A-38339).

3.4.2 Temperature conditions. The lubrication and overload selection shall permit operation of the compressors without damage in ambient temperatures ranging from 0° to +160°F for classes 1, 2, and 3 compressors, and from -25° to +160°F for classes 4 and 5 compressors. All compressors shall be capable of withstanding storage temperatures ranging from -65° to +160°F. This compressor suction cavity shall be capable of withstanding without damage, instantaneous pressure changes from the pressures corresponding to the minimum saturated evaporator temperature to the maximum saturated discharge temperature specified in 3.4.1.

3.4.3 Capacity and power requirements. The compressors specified herein shall have the capacity and power requirements specified in table II.

TABLE II. Compressor Capacities and Power Requirements

CLASS	TYPE	CAPACITY (MINIMUM)	BTU/HR (MAXIMUM)	BTU/WATT/HR (MINIMUM)	POWER FACTOR (MINIMUM)	ALLOW. POWER (KW) (MAXIMUM)
1	I	7,400	8,600	5.7	.85	1.4
	II	7,400	8,600	5.3	.59	1.6
2	I	11,000	13,100	5.4	.85	2.3
	II	11,000	13,100	5.8	.85	2.1
	III	11,000	13,100	6.0	.85	2.0
	IV	11,000	13,100	5.5	.60	2.2
3	I	22,200	26,000	6.0	.85	4.2
	II	22,200	26,000	6.0	.85	4.2
	III	22,200	26,000	5.7	.60	4.4
4	I	41,500	49,000	5.9	.85	7.5
	II	41,500	49,000	5.7	.60	8.0
5	I	65,000	76,000	5.9	.85	12.0
	II	65,000	76,000	5.7	.60	13.0

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3.4.3.1 The ratings specified in table II shall apply with the compressor operating at the following conditions:

Ambient air temperature blowing over compressor at approximately

600 fpm	150°F
Saturated discharge temperature	150°F
Evaporator temperature	50°F
Subcooling temperature	10°F
Suction superheat	15°F

3.4.4 Low temperature operation. Crankcase heaters shall be provided on classes 1, 2, and 3 compressors if required to prevent undesirable migration of liquid refrigerant to the compressor during shutdown or high starting torque conditions. Classes 4 and 5 compressors shall be provided with crankcase heaters to permit starting at -25°F ambient conditions.

3.4.5 Electric motors. The motor rated voltage phase and frequency shall be as specified in 1.2.

3.4.5.1 Starting torque. The starting torque requirements of the compressors shall be low, as the compressor will always start with equalized high and low side pressures. Kits for reluctance starting (run capacitor, start capacitor, and switching relay) shall be provided on all dual 50/60-Hz, single-phase compressors when starting torque is marginal.

3.4.5.2 Variation from rated voltage. The compressor shall start and operate as specified in NEMA Standard MG-1, paragraph entitled Variation From Rated Voltage, with voltage variations as specified for induction motors.

3.4.5.3 Variation from rated frequency. The compressor shall start and operate as specified in NEMA Standard MG-1, paragraph entitled Variation From Rated Frequency.

3.4.5.4 Combined variation of voltage and frequency. The compressor shall start and operate as specified in NEMA Standard MG-1, paragraph entitled Combined Variation Of Voltage And Frequency, except the combined variation tests shall be conducted with the compressor operating at rated conditions as specified in 3.4.3 herein.

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3.4.5.5 Motor protection (see 6.3.3 and 6.3.4). The compressor motor shall be equipped with a combination protector. The current protector shall be located remotely from the compressor and shall sense the current in all ungrounded power leads of the 3-phase compressor and at least one current-carrying line in single-phase compressors. The current sensor shall be required to break the above current-carrying lines upon current overload. In addition, the current-sensing devices shall contain a set of auxiliary contacts connected into a direct-current control circuit for the air conditioner. The auxiliary contacts shall open with the current sensor on compressor current overload; thereby interrupting the direct-current control circuit and breaking all current-carrying lines to the compressor. The auxiliary contacts shall reset automatically when the current sensor is manually reset. The auxiliary contacts shall be sized for compatibility with the control circuit. The current-sensing device shall have inherent temperature-compensation features and shall be adequately sized to prevent nuisance failures. A moistureproof, transparent, flexible seal shall be provided over the reset switch and shall in no way restrict the movement or prevent resetting the switch.

3.4.5.6 Electrical connections. Each compressor shall be provided with an electrical connector of the lightweight, environmental MS series. Connections shall conform to the wiring diagrams of the specific air conditioner (see 1.2). The MS connectors shall be as follows:

Compressor Class	Connector
1	MS3102-16S-1P
2 and 3	MS3102-20-15P
4 and 5	MS3102-24-11P

3.4.6 Environmental exposure. The compressors shall conform to all requirements specified herein after being exposed to the following environmental conditions:

a. Salt-water sea atmosphere

b. Fungus growth as encountered in tropical climates

c. Sand and dust particles as encountered in desert areas

d. Rain as encountered in any locale

e. Humidity.

3.4.6.1 Vibration. The compressors shall successfully operate after being subjected to vibration as specified in test method 514 of MIL-STD-810 for category h equipment.

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3.4.6.1.1 Vibration transmittance. The compressors shall be so designed that less than +0.1g acceleration (at any frequency) will be transmitted to the adjoining structure.

3.4.6.2 Shock. The compressors shall satisfactorily operate without impaired performance or breakage after being subjected to shock as specified in procedures I and II of method 516 of MIL-STD-810.

3.4.7 Weight, dimensions, connection sizes, and mounting. The weight of each compressor shall not exceed the requirements specified in table III. (The compressor plan view projection dimensions and heights are overall, and are the maximum allowable.) The compressors shall be provided with sweat-type refrigerant suction and discharge fittings suitable for use with copper tubing, and of the sizes listed in table III. A minimum length corresponding to the tubing diameter shall be provided for attachment of copper tubing to the mating sweat fitting. Mounting dimensions shall be as specified in table III, and the specific air conditioner as listed in MIL-A-38339 (see 1.2). The location and orientation of connections, overall configurations, and mounting shall permit interchangeability of alternate compressors.

TABLE III

CLASS	TYPE	MTG DIM (INCHES-NOM)	SUCTION CONN (INCHES-NOM) IDF / ODM	DISCH CONN (INCHES-NOM) IDF / ODM	MAX COMPL WT. (LBS[1])	MAX PROJ. VIEW PROJ. DIM (INCHES)	MAX OA HEIGHT (INCHES)
1	All	4.81X8.00	0.500	0.375	50	9.25X9.75	12.25
2	All	4.81X8.00	0.500	0.375	50	9.25X9.75	12.25
3	I and II	7.50X7.50	0.750	0.500	87	10.50X12.50	16.56
	III	7.50X7.50	0.750	0.500	76	10.50X12.50	16.56
4	I	7.50X7.50	0.875	0.625	100	13.50X12.88	17.00
	II	7.50X7.50	0.875	0.625	93	13.50X12.88	17.00
5	I	8.50X8.50	1.125	0.625	125	14.75X12.75	17.50
	II	8.50X8.50	1.125	0.625	120	14.75X12.75	17.50

[1] Compressor weight shall include complete compressor with oil charge, oil separator, and an oil cooler (if required).

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3.5 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.6 Electromagnetic interference. The compressors shall conform to the electromagnetic interference requirements as specified for class Am equipment in MIL-STD-826 when assembled into the air conditioner.

3.7 Acoustic noise level. The noise level of the hermetic compressors when operating in the applicable air conditioner (MIL-A-38339) shall not cause the air conditioner to exceed the noise level limits specified in the applicable detail specification when tested in accordance with the noise level test of MIL-A-38339.

3.8 Operational markings

3.8.1 Wiring diagram. A wiring diagram which incorporates all circuits from the power source to the motor shall be furnished. This diagram may be affixed to the compressor on the inside of the control box. If affixed to the compressor, it shall be inscribed on a corrosion-resistant plate.

3.8.2 Nameplate data. The following data shall be incorporated in a nameplate made of corrosion-resistant material or corrosion-protected material and securely attached to the compressor in a conspicuous place:

Refrigerant _____
Manufacturer's name _____
Manufacturer's part No. _____
Serial No. _____
Specification No. _____
Class - Type _____
Volts, Phase _____
Frequency (Hz) _____
Full load current _____ (Amps)
Lock Rotor Current _____ (Amps)
Ult. Trip Current _____ (Amps) (71°C)
Model of Motor Protectors _____
Contract No. _____
Type and Recommended Oil _____
Oil Charge - (Pints) _____

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.10 Workmanship. The compressors shall be fabricated, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from defects, burrs, and sharp edges; accuracy of dimensions, thoroughness of soldering, welding, brazing, and riveting; and alignment of parts and tightness of assembly screws, bolts, et cetera.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the compressors shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions. The test conditions shall be as described under the tests to which they apply.

4.3.1 Readings. Readings of all pertinent pressures, temperatures, rotational speeds, and other necessary data required to demonstrate compliance with applicable portions of section 3 shall be recorded at intervals not exceeding 30 minutes during all tests specified herein.

4.3.2 Post-test inspection. Upon completion of all testing or when deemed necessary by the contracting officer, the compressor may be removed from the shell of a welded assembly or the covers removed on a field service type for physical examination.

4.4 Preproduction testing (see 6.2)

4.4.1 Test samples. The test samples shall consist of models representative of the production equipment. Unless otherwise specified (see 6.2), two prototypes of each type compressor required by the contract shall be subjected to the preproduction tests specified in 4.4.3.

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4.4.2 Test report. Upon completion of the preproduction tests, a preproduction test report in accordance with MIL-STD-831 shall be prepared and two complete copies and one reproducible copy of the report furnished to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the preproduction test report (same quality and quantity of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Summary of the engineering analysis and of any tests conducted to determine assignable cause for all failures and irregular operations identified.

c. Summary of the engineering analysis behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on use or maintenance conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

4.4.2.2 Preproduction test approval. All compressors that have been successfully tested and have had preproduction test approval to this specification must have the compressor motors identified by the manufacturer's name and part number as it appears on the motor. Any preproduction test approval given to a particular compressor shall apply only for the motor part number used during the original preproduction testing. The same compressor with a different motor shall have to be retested in accordance with the preproduction tests of this specification to be accepted.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6. When two samples are used, they shall be allocated to the preproduction tests as follows and tested as described under 4.6:

Sample No. 1

Examination of product	See 4.6.1
Storage test	See 4.6.3
Environmental tests	See 4.6.4

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High potential test	See 4.6.5
Capacity tests	See 4.6.6
Vibration transmittance test	See 4.6.12
Vibration test	See 4.6.13
Shock test	See 4.6.14
Endurance test	See 4.6.16.

Sample No. 2

Examination of product	See 4.6.1
Tilted position test	See 4.6.9
Winding temperature test	See 4.6.7
Locked rotor test.	See 4.6.8
Electromagnetic interference test	See 4.6.11
Noise level test	See 4.6.10
Pressure tests	See 4.6.15.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each compressor shall be subjected to the following tests as described under 4.6:

a. Examination of product	See 4.6.1
b. Capacity verification (except that the test run shall be of 30 minutes duration)	See 4.6.2

4.6 Test methods

4.6.1 Examination of product. The compressor shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, markings, and finishes.

4.6.2 Capacity verification test. A capacity test at 50°F saturated suction temperature, 150°F saturated discharge temperature, and 65°F actual suction temperature shall be conducted in accordance with 4.6.6. Test runs shall be of 30 minutes duration. The resistance to ground shall be recorded. Winding temperatures in excess of motor manufacturer's guaranteed limits or automatic trip-points at less than motor manufacturer's limits or for reasons other than motor protection shall be cause for rejection.

4.6.3 Storage tests

4.6.3.1 High temperature test. The compressor (with normal operating oil and refrigerant -22 charge) shall be subjected to the high temperature test specified in procedure I of MIL-STD-810, method 501. Both the high- and

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low-pressure cavities of the compressor shall be subjected to liquid refrigerant -22 pressures corresponding to the saturated temperatures throughout exposure. The compressor shall be satisfactorily operated upon conclusion of this test.

4.6.3.2 Low temperature test. The compressor shall be subjected to the low temperature test specified in procedure I of MIL-STD-810, method 502, except the exposure temperature shall be -65°F and maintained for a period of not less than 48 hours. The compressor shall be satisfactorily operated upon conclusion of this test.

4.6.4 Environmental tests. Upon completion of the storage tests specified in 4.6.3.1 and 4.6.3.2, the following tests shall be conducted in accordance with the specified procedures of MIL-STD-810:

Test	Procedure
Sand and dust	Procedure I, method 510
Salt-fog	Procedure I, method 509
Fungus resistance	Procedure I, method 508
Rain	Procedure I, method 506
Humidity	Procedure I, method 507

4.6.5 High potential test. The compressor electrical circuit shall be subjected to and shall satisfactorily complete a high potential test in accordance with NEMA Standard MG-1, paragraph entitled High Potential Test.

4.6.6 Capacity tests. Capacity tests shall be conducted in accordance with ASHRAE Standard 23, as modified to meet the temperatures and pressures specified herein and shall be run prior to the endurance test. The results of capacity and wattage input from the operational tests specified in 4.6.6.2 (runs 1 and 10) shall not vary more than 3 percent. In addition to the data required by 4.3.1, the following shall be recorded:

a. Capacity (BTU/hr)

b. Voltage and frequency

c. Amperes

d. Watts input

e. Power factor

f. Winding temperature (resistance method of Seely circuit tester).

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4.6.6.1 Test conditions. The following conditions shall apply for all capacity tests:

a. Voltage variation (±2V)

b. Frequency variations (±1/2 cycle)

c. Subcooling (10°F)

d. Superheat at compressor (15°F)

e. Ambient airflow over compressor (600 fpm).

4.6.6.2 Operational tests. Capacity test runs shall be performed under the conditions specified in table IV and shall be conducted in sequential order from 1 to 10. The compressor shall be mounted in a manner simulating that under actual service. Each test run shall be of 5 hours duration.

TABLE IV. Capacity Test Run Conditions

Run	Saturated Discharge Temp °F	Ambient Temp °F	Saturated Suct Temp °F
1	120	120	30
2	130	130	30
3	130	130	60
4	150	150	30
5	150	150	50
6	150	150	60
7	160	160	40
8	160	160	60
9	60	20	20
10	120	120	30

4.6.7 Winding temperature test. The winding temperatures shall be measured by the resistance method. The compressor shall be operated at 150°F saturated discharge temperature, 50°F saturated suction temperature, and 150°F ambient air temperature across the compressor. The temperature of the suction gas entering the compressor shall be increased in 5°F increments starting at 60°F. The winding temperature shall be recorded for each suction gas temperature increment. The period of operation for each increment shall be of

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sufficient length for stabilized winding temperature. The test shall be continued until the maximum winding temperature, as recommended by the manufacturer, is reached or the compressor overtemperature protector opens the power circuit. The foregoing tests shall be conducted at rated voltage and frequency.

4.6.8 Locked rotor test. The locked rotor test shall be conducted in accordance with Underwriters' Laboratories, Incorporated Standard No. 303 paragraph entitled Locked Rotor Test - Hermetic Compressor Motors.

4.6.9 Tilted position test. The compressor shall be operated while tilted 10 degrees from the horizontal position in each of the following directions: front, back, and each side. The compressor shall operate for 4 hours at rated conditions in accordance with 4.6.6 in each of the four specified positions.

4.6.10 Acoustic noise level test. The noise levels produced by the appropriate air conditioner, when the applicable compressor of the detail specification is installed, shall not exceed the sound pressure level limits of the detail specification when tested in accordance with the noise level test of MIL-A-38339.

4.6.11 Electromagnetic interference test. When installed in the air conditioner, the compressor shall assure compliance with the applicable electromagnetic interference test requirements of MIL-STD-826, class Am, as follows: Test methods 1001, 2001, 3001, 3002, and 4001. Short duration (transient) interference requirements of test methods 3002 and 4001, are relaxed 20 db for the air conditioners.

4.6.12 Vibration transmittance test. A vibration survey of each test run of 4.6.6.2 shall be conducted. Transmitted acceleration (at any frequency) in excess of $\pm$0.1g shall be cause for rejection.

4.6.13 Vibration test. A compressor mounting bracket shall be fabricated and attached to a rigid base. The compressor shall be mounted on the bracket using 70 durometer vibration washers similar to those covered by Industrial Rubber Goods Company part No. D-4525-265. The compressor shall satisfactorily operate after being subjected to a vibration test conducted in accordance with MIL-STD-810, test method 514, equipment category h, procedure X and curve AB. The resonant search and resonant dwell shall be at 1-1/4g input, and the sinusoidal cycling test shall be at 2-1/2g input. The time used for resonant search shall not be counted in total test time.

4.6.14 Shock test. The compressor shall satisfactorily operate after being subjected to a shock test conducted in accordance with MIL-STD-810, method 516, procedure I with a shock wave in accordance with figure 516-1 (terminal-peak sawtooth) with an amplitude of (b) and a time duration of (d) as listed for ground equipment, and procedure II except that the compressor shall be

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dropped on each face and each corner and each edge from a height of 30 inches for class 1 through class 4 compressors and a height of 21 inches for the class 5 compressor.

4.6.15 Pressure tests. Pressure tests shall be performed on and applied internally to the compressor crankcase under the conditions specified in table V. The capacity verification test specified in 4.6.2 shall be conducted after both gas tests, except that the test run shall be of 30 minutes duration. Failure to pass the capacity verification test or rupture at pressure less than 675 psig shall constitute failure of this test.

TABLE V. Conditions For Pressure Tests

Test	Cycles	Pressure Range (PSIG)	Minimum Pressure Increase Time Rate
Gas	15	0 to 434	5 pounds/second
Gas	24	0 to 380	Instantaneous
Rupture (Hydrostatic)	1	0 to 675	5 pounds/second

4.6.16 Endurance test. An endurance test shall be conducted on compressors which have successfully passed the tests specified in 4.6.6. The airflow across the compressor during the endurance testing shall be approximately 600 fpm. The ambient temperature throughout the endurance test shall be the same as the specified saturated discharge temperatures. The refrigerant oil shall withstand the discharge temperatures of the compressor for the duration of the life test.

4.6.16.1 Test schedule. The schedule specified in table VI shall be followed in performing the endurance tests. Resistance to ground shall be recorded after each 100 hours of operation or each test, whichever is the less time.

4.6.16.2 Test loop. The compressor shall be tested on a refrigerating test loop of sufficient sizing for each class of compressor tested.

4.6.16.3 Test installation. When installed in the appropriate air conditioner (MIL-A-38339), the compressors shall satisfactorily pass the Endurance tests of MIL-A-38339, except for the Heating test.

4.6.17 Reliability demonstration. Satisfactory completion of all tests specified herein without failure demonstrates compliance with the quantitative reliability requirements.

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TABLE VI. Endurance Test Schedule

RUN 1. 250 hours of intermittent operation under the following conditions:

15 minutes on
5 minutes off

a. 150°F saturated discharge temperature

b. Suction gas temperature, 25°F superheat

c. 150°F ambient temperature

d. Evaporation temperature shall be varied periodically with approximately one-third of the total 250 hours being spent at each of the saturated suction temperatures of 40°, 50°, and 60°F.

RUN	HOURS	VOLTS	HERTZ	SATURATED SUCTION TEMP °F	SATURATED DISCHARGE TEMP °F	ACTUAL SUCTION TEMP °F
2	16	Low (5%)	Rated	50	150	75
3	16	High (10%)	Rated	50	150	75
4	16	Rated	Low (5%)	50	150	75
5	16	Rated	High (5%)	50	150	75
6	16	Low (5%)	Rated	40	150	65
7	16	High (10%)	Rated	40	150	65
8	16	Rated	Low (5%)	40	150	65
9	16	Rated	High (5%)	40	150	65
10	16	Low (5%)	Rated	45	160	70
11	16	High (10%)	Rated	45	160	70
12	140	Rated	Rated	35	150	60
13	140	Rated	Rated	40	150	65
14	150	Rated	Rated	45	150	70
15	150	Rated	Rated	50	150	75
16	500	Rated	Rated	30	100	55
17	500	Rated	Rated	40	125	65
18	500	Rated	Rated	50	150	75
19	250	Rated	Rated	60	160	85
20	250	Rated	Rated	20	60	45

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4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall level A or C as specified (see 6.2).

5.1.1 Level A. Compressors shall be preserved and packaged in accordance with method III of MIL-P-116. All openings of the electrical connection box shall be sealed with class 1 tape conforming to PPP-T-60.

5.1.2 Level C. Preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each compressor, preserved and packaged as specified in 5.1.1, shall be packed in an overseas-type style B container conforming to PPP-B-601. Containers shall be of uniform shape and size and minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-601.

5.2.2 Level B. Level B packing shall be the same as specified for level A, except that containers shall be of the domestic type, style A.

5.2.3 Level C. Compressors, preserved and packaged as specified in 5.1.2, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. Internal blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. The free-fall drop test shall be in accordance with appendix A of MIL-STD-1186. The test shall not be required when level C packing is specified.

5.4 Marking. In addition to any special marking required by the contract or order (see 6.2), the interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The hermetic compressors covered by this specification are for use in combination with other standard refrigeration components in air conditioning systems which employ refrigerant-22.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Class and type of compressor required (see 1.2)

c. Point of inspection and quantity of preproduction samples (see 4.4 and 4.4.1)

d. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2)

e. Special shipment marking (see 5.3)

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence Level = 90 Percent

$$MTBF = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x}$$

$$\text{where } x = \frac{\text{Mission time}}{MTBF}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

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6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.3.3 Combination protector. Combination protector is defined as a protective device which is responsive to motor temperature and current and which protects the motor from overheating due to overload or failure to start by automatically disconnecting the motor from the source of power supply.

6.3.4 Inherent motor protection. A hermetic compressor is considered to have inherent motor protection when it is equipped with a combination protector applied to the motor in such a manner that it protects the motor from overheating due to reduction or loss of cooling medium, overload, failure to start, or phase failure by automatically disconnecting the motor from the source of power supply.

6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

BTU/hr.	British thermal units per hour
db	decibels
fpm	feet per minute
g	gravitational unit
Hz	Hertz, cycles per second
kw	kilowatt
MTBF	mean-time-between-failure
NEMA	National Electrical Manufacturers Association
psig	pounds per square inch gage
Ult	ultimate
V	volt

6.5 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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